

FLEA NEWS 34

Department of Entomology
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Ames, Iowa 50011 U.S.A.

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Compiled by Robert E. Lewis and Joanne H. Lewis

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Please find herewith:

- lists of additional literature for 1975 (list 21), 1984 (list 7), 1985 (list 5), 1986 (list 3), and 1987 (list 1).
- additions and changes to the mailing list

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MILAN JURÍK

14-VI-1930 19-X-1986

Dr. Alexander Dudich has written to inform us of the death of Ing. Milan Jurík, in Brno, after a long illness. Subsequently Drs. D. Cyprich and J. Ryba provided the following obituary.

Ing. Milan Jurík was born in Vinosady, near Bratislava, Czechoslovakia, where his father was an educator. Upon completing his secondary education he entered the Agricultural University at Brno where, from 1949 to 1953, he studied in the Faculty of Forestry. In 1952 he was appointed to a teaching assistantship in the Department of Zoology. Later he was appointed lecturer in the Department of Veterinary Sciences. In 1971 he returned to the Department of Zoology, and in 1977 he was promoted to the rank of associate professor.

During his early years at the university Jurík specialized in mammalogy. As his career progressed he devoted more and more of his time to ectoparasites, with particular emphasis on the species composition and ecology of bird fleas. He also studied the problem of fleas and their control in the breeding farms of fur-bearing animals, and in later years concentrated on studies of the fleas associated with moles in Czechoslovakia. The results of these later studies remain unpublished.

Milan Jurík was both a scientist and a teacher and was active both at the University and in public life. His unselfishness in assistance to his students and colleagues, his meticulousness in his work and his kind words will be missed by all who knew him.

For further biographical information the reader is referred to: Rosický, B., 1980. Fiftieth birthday of Ing. M. Jurík CSc. Folia Parasitol., Praha 27: 287-288.

More than half of Jurík's publications were devoted to fleas. Following is a bibliography of these studies compiled by Dr. Dusan Cyprich.

MILAN JURÍK

- 1955 - Príspevok k poznaniu blch (Aphaniptera) ČSR. (A contribution to the knowledge of fleas (Aphaniptera) of ČSR.) Acta Univ. Agric. Brno (2): 176-180.
- 1957a - Ektoparazity (Anoplura a Aphaniptera) drobných cicavcov Jeseníkov. (Ectoparasites (Anoplura and Aphaniptera) of the small mammals of Jeseníky mountains.) Acta rerum nat. Dist. ostraviensis 18: 112-125.
- 1957b - Zpráva o stúdiu ektoparazitov Eliomys quercinus (L.) a Glis glis (L.). (Report on study of ectoparasites of Eliomys quercinus (L.) and Glis glis (L.).) Biológia (Bratislava) 12: 18-28.
- 1960 - Aphaniptera drobných cicavcov Badínského pralesa (Kremnické hory). (Aphaniptera of small mammals in Badínsky forest (Kremnické mountains.) Biológia (Bratislava) 15: 849.
- 1962 - Príspevok k poznaniu Aphaniptera ČSSR II (Nosopsyllus Jordan, 1933). (A contribution to the knowledge of Aphaniptera of ČSSR II (Nosopsyllus Jordan, 1933).) Biológia (Bratislava) 17: 383-388.
- 1966 - The Aphaniptera of the South Slovakian karst. (A contribution to regional parasitology.) Acta Univ. Agric. Brno (1): 103-110.
- 1967 - Aphaniptera vtáčích hniezd. (Aphaniptera of birds' nests.) Thesis. 192 pp., 21 figs.
- 1968a - Ceratophyllus rossittensis Dampf, 1913 in Czechoslovakia. Folia parasitol. (Praha) 15: 349-350.
- 1968b - Fleas of the mole Talpa europaea L. in Czechoslovakia (Aphaniptera). Acta ent. bohemoslov. 65: 67-75.
- 1969 - Beschreibung und Verwendung einer Neukostruktion des Thermoeclectors. Ent. Nachrichten 13: 95-98.
- 1970 - The distribution of specific and nonspecific parasites (Siphonaptera). Presented at the II. Congress of Parasitology, Washington, D.C., September 8, 1970.
- 1971 - Počet generácií vtáčích druhov blch v prebiehu roka. (Number of generations of bird fleas during the year.) Zprávy cs. spol. zool. (1-3): 102-104.
- 1972 - The distribution of specific and nonspecific parasites (Siphonaptera). J. med. Entomol. 9: 605-606.
- 1973 - Niektoré poznatky o možnostiach adaptácie blch (Siphonaptera) na nového hostiteľa. (Some knowledge on possibilities of adaptation of fleas (Siphonaptera) to the new host. IV. Čelost. sjezd čs. zool., Č. Budějovice 9-15.IX.1973.
- 1974a - Nové poznatky o biologii blech. (New knowledge about the biology of fleas.) Scripta medica (Brno) 47: 100-101.
- 1974b - Bionomics of fleas in bird nests in the territory of Czechoslovakia. Acta Sci. nat. Brno 8: 1-54.
- 1975 - Quantitative evaluation of the classification of fleas as bird parasites. Acta Univ. Agric. Brno 23: 555-566.
- 1976a - Interrelationships between studied zones of avian Aphaniptera and linkages of these to zones of mammalian Aphaniptera. Acta Univ. Agric. Brno 24: 161-169.

- 1976b - Monopsyllus sciurorum (Schrank, 1781) ako parazit v chovoch noriek v ČSSR. (Monopsyllus sciurorum (Schrank, 1781) as a parasite in breeding farms of minks in ČSSR.) Zprávy čs. spol. zool. (7-9): 11-14.
- 1978 - Significance of individual bird species as hosts of fleas (Siphonaptera). Acta Univ. Agric. Brno 26: 111-116.
- 1981 - Podmienky umožňujúce adaptáciu parazita (blachy) na nového hostiteľa. (Conditions allowing an adaptation of parasites (fleas) on a new host.) Celost. zool. konf., Bratislava, 24-28.VIII.1981, pp. 304-307.
- 1983a - Ceratophyllus vagabundus insularis Rothschild, 1906 and Ceratophyllus rossittensis Dampf, 1913 (Siphonaptera) in Czechoslovakia. Folia parasitol. (Praha) 30: 169-173.
- 1983b - To the knowledge of ecological conditions affecting the occurrence of specific and non-specific flea species on their hosts (Talpa europea-Siphonaptera). Biológia (Bratislava) 38: 949-957.
- 1984a - Ekologické problémy československej siphonapterologie. (Ecological problems of czechoslovak siphonapterology.) Práce Slov. ent. spol. 5: 55-65.
- 1984b - Pásmo siphonaptérií synantropných hlodavcov. (Zone of Siphonaptera of synanthropic rodents.) Práce Slov. ent. spol. 5: 133-140.
- with F. Kukla, 1970 - Výskyt blech na norčích farmách. (The occurrence of fleas on mink farms.) Chovatel 9: 204-205.
- with F. Kukla, 1974 - The occurrence of fleas (Siphonaptera) on mink farms in Czechoslovakia. Acta Univ. Agric., Brno 22: 131-139.
- with A. Dudich & I. Grulich, 1984 - Palaeopsylla aff. cisalpina Jordan et Rothschild, 1920 (Siphonaptera:Hystrichopsyllidae) in ČSSR. Biologia (Bratislava) 39: 205-213.
- with D. Cyprich & M. Krumpál, in press - Predbežná správa o výskyte blch (Siphonaptera) v hniezdach vrabcov (Passer ssp.) na Slovensku a Morave. (Preliminary report of the occurrence of fleas (Siphonaptera) in Slovakia and Moravia. Práce Slov. ent. spol.

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DESCRIPTIONS PER ANNUM

Among the other useful bits of information included in the 20 issues of Flea News by Frans Smit, the founder of this newsletter, were tables indicating the numbers of species and subspecies described annually since 1758. The first of these appeared on page 2 of FN 5. (If numbered consecutively as the issues appeared, this would be page 34.) An update of this table appeared on page 1 of FN 17 (page 171). The latter included the years 1974-1979. Below is a supplement to these lists incorporating a correction for 1965 and extending to the end of 1986.

<u>Year</u>	<u>Valid</u>	<u>(Synonyms)</u>	<u>Total</u>	<u>(+Synonyms)</u>
1965	33	(3)	1796	2197
1966	33	-	1829	2230
1967	26	(2)	1855	2258
1968	20	(1)	1875	2279
1969	41	(1)	1916	2321

1970	14	(3)	1930	2338
1971	31	(2)	1961	2371
1972	32	(1)	1993	2404
1973	28	(1)	2021	2433
1974	29	(1)	2050	2463
1975	53	(2)	2103	2518
1976	28	-	2131	2546
1977	47	(4)	2178	2597
1978	29	(1)	2207	2627
1979	49	(3)	2256	2679
1980	24	-	2280	2703
1981	25	-	2305	2728
1982	29	-	2334	2757
1983	14	-	2348	2771
1984	14	-	2362	2785
1985	17	-	2379	2802
1986	3	-	2382	2805

PUBLICATIONS PER ANNUM

Also, on pages 1 and 2 of FN 19 (pages 185-186) is a table listing the number of flea publications per annum beginning with the year 1475 and ending with 1979. Since some retrospective additions have been made, below is an update of Smit's table from 1972 through 1986. Figures are based on citations in Flea News for this period.

1972 + 240 = 9249	1980 + 287 = 11437
1973 + 208 = 9457	1981 + 310 = 11747
1974 + 376 = 9833	1982 + 400 = 12147
1975 + 363 = 10196	1983 + 256 = 12403
1976 + 222 = 10418	1984 + 249 = 12652
1977 + 247 = 10665	1985 + 198 = 12850
1978 + 221 = 10886	1986 + 83 = 12933
1979 + 264 = 11150	

Bear in mind that these are minimal numbers since we make no concerted effort to search the veterinary and public health literature.

---R.E.L.

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A NEW JOURNAL, Medical and Veterinary Entomology, has been initiated by the Royal Entomological Society of London. Published for the RESL, it will appear quarterly at an annual subscription price for 1987 of £ 58.50(UK), \$133.00(USA, Canada and Japan) and £ 77.00(elsewhere), post free. In the editorial preface the editors state: "Medical and Veterinary Entomology will cover all aspects of the biology and control of arthropods that attack man and animals. This journal will carry original research papers on insects and other arthropods of medical and veterinary importance as endoparasites, ectoparasites, allergens, pests or vectors of pathogens. Control methods of all types will be reported at both the research and operational levels. The main subjects of interest will be haematophagous arthropods, but all kinds of synanthropic and zoophilic arthropod biology will be considered."

Volume 1, No. 1, 108 pages, 6 articles on mosquitoes, 3 on muscoid flies, 2 on ticks, and 1 each on ceratopogonids (sandflies) and simuliids (black flies).

Information concerning subscriptions, back numbers, offprints, etc. may be obtained from Blackwell Scientific Publications, P.O. Box 88, Oxford, U.K.

---R.E.L.

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A REQUEST FOR COLLECTORS AND SYSTEMATISTS - Current studies of the male genitalia in pulicid fleas have revealed the great utility of scanning electron microscopy in the interpretation of structure and in comparative anatomy. In some cases it has proven possible to prepare slide mounted specimens for such examination, but in general the lack of uncleared, unmounted material has been frustrating.

Since scanning electron microscopes have become very common, and relatively easy to use, we make the following suggestion: Collectors and curators should, routinely and whenever possible, in preparing fleas for study, retain at least some specimens uncleared and stored in 70% alcohol. If identification is uncertain and examination of slide mounted material is necessary, the specimens may be cleared in xylene and temporary mounts made in alpha terpineol. This usually provides enough clearing for the examination of most chitinous structures. The specimens may then be dehydrated in ethanol and stored, or prepared for scanning without undue damage. Metal coated specimens, mounted on small copper disks in any of the standard ways, may be stored indefinitely and do not take up much more room than slide mounted material. The general availability of material prepared in this way could be a great boon to students of the order.

---T. B. Cheetham

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CONGRATULATIONS - THOMAS B. CHEETHAM received the Doctor of Philosophy degree from Iowa State University on 16 May 1987. His dissertation is entitled "A comparative study of the male genitalia in the Pulicoidea". BETH A. SCHRAMM successfully defended her dissertation, "A taxonomic revision of the genus Plocopsylla Jordan, 1931 (Siphonaptera: Stephanocircidae)", in June 1987. Her Ph.D. degree will be awarded at the August commencement exercises at Iowa State University. Both students worked under the direction of Dr. Robert E. Lewis.

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PLAGUE UPDATE - According to recent numbers of the Technical Information Bulletin of the Armed Forces Pest Management Board, the following cases of plague have been reported. Two non-fatal cases were reported from Madagascar during November of 1986. In Brazil the "epidemic" reported in October in Paraiba State was said to be threatening neighboring states in November with 40 to 50 acknowledged cases, 6 of which were fatal. Forty-five cases were also reported from Franz Tamayo Province, Bolivia, between 7-February and 19-March-1987.

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FLEA TRIVIA QUESTION: What was the only year in this century in which no new flea species were described? Answer in FN 35!

NOTES

The editors would like to apologize for the confused pagination in the bibliographic section of Flea News 33. The transposition of pages 233 and 237 was accomplished at the copy center and was not discovered by us until after the issue was in the mail.

Overlooked in the rush to produce Flea News 33 was a note from Frans Smit to the effect that he had just completed going over the proofs for Volume VII of the Catalogue of the Rothschild Collection of Fleas. We have since been informed by the Director of Marketing of the Oxford University Press, that they, in collaboration with the British Museum (Natural History), will be publishing the following title later this year:

Smit, F. G. A. M. Illustrated Catalogue of the Rothschild Collection of Fleas. Volume VII. Rhopalopsyllidae.

No further particulars are available at this time but they have requested a copy of the Flea News mailing list.

The annual conference of the SOCIETY OF VECTOR ECOLOGISTS will be held at the Asilomar Conference Center, Monterey, California, on November 18, 19 and 20, 1987. For further information contact B. Fred Beams, Chairman, Public Information Committee, The Society of Vector Ecologists, Inc., P.O. Box 87, Santa Ana, California 92702 U.S.A. Telephone: (714) 971-2421.

A recent advertisement of books available from THOMSON PUBLICATIONS, P.O. Box 9335, Fresno, California 93791 U.S.A., includes one by Walter Weber, "Fleas, Ticks and Cockroaches - Disease Transmitters". Soft-cover, 42 pp., price \$10.00(US). The flyer says of the book, in part, "...This book brings together....information on the vectors and the diseases in a condensed form in layman's language." The year of publication is not given.

The editors wish to take this opportunity to thank all of those colleagues who responded to our request for literature. A large percentage of our needs was filled by Ms Annette K. Thomas of the Entomology Department of the British Museum (Natural History), to whom we extend special thanks. Publications were also provided by J.C. Beaucournu, A. Benton, A. Dudich, L. Guimarães, M. Hastriter, E. W. Jameson, J. Lane, J. Máca, J. C. Morales, J. Segerman, W. Skuratowicz and N. Wilson to all of whom we extend our sincere appreciation.

* * *

REPRINTS RECEIVED - We also wish to thank the following for supplying reprints and citations since the publication of Flea News 33:

W. Cudmore	R. S. George	R. Pilgrim	T. G. Schwan
B. Drucker	G. E. Haas	H. E. Ponce Ulloa	D. Q. Wang
L. A. Durden	H. Katz	V. Sapegina	B. Williams

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Juan Carlos Morales, Department of Wildlife and Fisheries Sciences, Texas A &
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Dr. F. J. Radovsky, Department of Entomology, Oregon State University,
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DELETIONS

D. A. Humphries
M. Jurík

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Our apology for the tardy issue of Flea News 34. With the North Central
Branch meetings of the Entomological Society of America in Des Moines and a
Cooperative State Research Service departmental review, in addition to our
regular duties, time has been of the essence this spring. Thus the late issue
of number 34.

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(1975)

A PREFATORY NOTE TO THE SPECIES LIST - In keeping with Mr. Smit's practice of numbering the taxa chronologically, the following list includes numbers for those taxa described in 1985. Readers are urged to call any omissions in these lists to our attention.

- 98501 australishaanxia Zhang & Liu Peromyscopsylla himalaica
- 98502 scaliforma Zhang & Liu Peromyscopsylla
- 98503 calceiforma Zhang & Liu Paradoxopsyllus
- 98504 conica Zhang, Wu & Liu Liuopsylla
- 98505 celebensis Lewis & Jones Sigmactenus
- 98506 hetera Lewis & Jones Gryphopsylla
- 98507 exoticus Lewis & Jones Stivalius
- 98508 orientalis Mahnert & Beaucournu Typhloceras poppei
- 98509 stenofronta Xie Rhadinopsylla (Actenophthalmus)
- 98510 rotunditruncata Cai & Liu Frontopsylla
- 98511 esinus Liu, Shi & Liu Typhlomyopsyllus
- 98512 eremita Beaucournu & Kowalski Caenopsylla
- 98513 xijiensis Zhang & Dang Dorcadia
- 98514 furcata Shi, Liu & Wu Macrostylophora
- 98515 qinghaiensis Zhang & Ma Ceratophyllus (Ceratophyllus)
- 98516 rhinopithecus Li Vermipsylla parallela
- 98517 postprojicia Liu, Qi & Li Frontopsylla (Orfrontia)

PROVISIONAL LISTS OF TAXA DESCRIBED IN 1986 AND 1987

1986

Coorilla allisoni Mardon
Frontopsylla (Frontopsylla) wagneri superjecta Zhang, Liu & Zhang
Paradoxopsyllus jinshajiangensis concavus Zhang, Liu & Zhang
Lentistivalius affinis Li
Plocopsylla diana Beaucournu, Gallardo & Launay
Neopsylla affinis deqinensis Xie & Li
Ctenophthalmus (Ctenophthalmus) bifidatus mancus Orhan & Beaucournu
Ctenophthalmus (Medioctenophthalmus) nifetodes tvrtkovici Brelih
Ctenophthalmus (Medioctenophthalmus) nifetodes petrovi Brelih
Ctenophthalmus (Medioctenophthalmus) nifetodes krystufeki Brelih
Ctenophthalmus (Medioctenophthalmus) nifetodes dzukici Brelih

1987

Caenopsylla janineae Beaucournu & Gouat

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LITERATURE ON SIPHONAPTERA PUBLISHED IN 1975 (list 21)

Van Veen, T. W. S. & A. N. Mohammad - Louse and flea infestations on small ruminants in the Zaria area. J. Niger. vet. Med. Assoc. 4(2): 93-96

(1984 - 1985)

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1984 (list 7)

- Arshad, M., G. A. Jan & Asmattullah Khattack - Insect fauna of birds' nests of N.W.F.P. Bull. Zool., Pakistan 2: 1-7
- Iqbal, Q. J. - Studies on the mating determinants in the rat flea *Nosopsyllus fasciatus* (Bosc). Bull. Zool., Pakistan 2: 27-31
- *Lakshminarayana, K. V., M. S. Mani & W. D. Eichler - On the relationship of the fossil flea *Saurophthirus longipes* Ponomarenko (Siphonaptera: Insecta). Rec. zool. Survey, India 81(3-4): 43-48
- O'Sullivan, H. M., C. M. Smal & J. S. Fairley - A study of parasite infestations in populations of small rodents (*Apodemus sylvaticus* and *Clethrionomys glareolus*) on Ross Island, Killarney. J. Life Sci., Royal Dublin Soc. 5(1): 29-42

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1985 (list 5)

- Ambros, M., A. Dudich, J. Kováčik & A. Štollmann - Ekroparazity (Acarina, Anoplura, Siphonaptera) mikromamáľí (Insectivora, Rodentia) Východoslovenskej nížiny. Zborn. Východoslov. Múz. v Košiciach, Prirod. Vedy 26: 127-157 [Journal also called Acta Musei Slovaciae Regionis Orientalis Košice]
- Anonymous - The Zeprox guide to flea control. Glaxovet Limited, Harefield, Uxbridge, Middlesex UB9 6LS, U.K. 10 pp.
- Beaucournu, J. C. & A. Aeschlimann - Une puce nouvelle pour la région Paléarctique occidentale: presence en Suisse de *Callopsylla gemina* (Ioff, 1946) (Siphonaptera, Ceratophyllidae). Bull. Soc. neuchatel. Sci. nat. 108: 55-62 [Erects new subgenus of *Callopsylla*, *Geminopsylla*, for *gemina* and *gypaetina*]
- Dudich, A. - Ektoparazitofauna drobných cicavcov (Insectivora, Rodentia) pohronskeho inovca (Západné Karpaty) 1. Siphonaptera. Rosalia (Nitra) 2: 247-264.
- Dudich, A. - Fleas (Siphonaptera) occurring in insectivores (Insectivora) and rodents (Rodentia) in the protected territory Pol'ana. Ochrana Prirody 6: 243-261
- *Griboedov, A. V. & G. P. Barabash - "Basic" somatic antigen or antigen obtained from *Yersinia pestis* by the method of Boivin-Mesrobianu. Zh. Mikrobiol., Epidemiol., Immunol. (3): 108-115
- *Hickin, N. - Pest animals in buildings, 2 world review. George Godwin, London, i-xiii, 1-385, illus.
- Hughes, R. G. - The relative efficacy of insecticidal collars containing propoxur or tetrachlorvinphos. Vet. Med. Rev., German Fed. Rep. (1): 80-88
- Obuch, J. - Contribution to the inventory research of vertebrates (Vertebrata) in the state nature reserve Suchy in the protected territory Malá Fatra. Ochrana Prirody 6: 215-230
- Saxena, V. K. - Some entomological observations in a suspected plague outbreak area of District Shimla (Himachal Pradesh). J. commun. Dis. 17(4): 344-346

(1985 - 1986)

- Sneath, P. H. A. & M. Stevens - A numerical taxonomic study of *Actinobacillus*, *Pasteurella* and *Yersinia*. J. gen. Microbiol. 131(10): 2711-2738
- Ugochukwu, E. I. & A. O. Apeh - Prevalence of ectoparasites of small ruminants in Nsukka, Nigeria. Intl. J. Zoonoses 12(4): 313-317
- Ugochukwu, E. I. & C. C. Nnadozie - Ectoparasitic infestation of dogs in Bendel State Nigeria. Intl. J. Zoonoses 12(4): 308-312

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1986 (list 3)

- Baungartner, D. L. & A. Kane - Wood ducks as accidental hosts of the Squirrel flea, *Orchopeas howardi* (Siphonaptera: Ceratophyllidae). Great Lakes Entomol. 19(4): 249-250
- Beaucournu, J. C., M. N. Gallardo & H. Launay - Pucés (Siphonaptera) nouvelles ou peu connues du Chili: Description de *Plocopsylla diana* n. sp. (Stephanocircidae). Ann. Parasitol. Hum. Comp. 61(3): 359-366 [Also describes male allotype of *Ctenoparia inopinata*]
- Brelih, S. - Ectoparasitical entomofauna of Yugoslav mammals. II. Siphonaptera from *Dinaromys bogdanovi* and *Chionomys nivalis* (Rodentia: Cricetidae). Scopolia No. 11: 1-47 [Ctenophthalmus (Medioctenophthalmus) nifetodes tvrtkovici, C. (M.) n. petrovi, C. (M.) n. krystufeki, C. (M.) n. dzukici]
- Briggs, O. M. - Flea control on pets in southern Africa. J. S. Afr. vet. Assoc. 57(1): 43-47
- Burton, H. W. & J. H. Barrett - External parasites of poultry. Queensland Agric. J. 112(1): 5-11 [Siph.: 8-10]
- Collart, M. G. & W. F. Hink - Sublethal effects of D-limonene on the cat flea, *Ctenocephalides felis*. Ent. Exp. Applic. 42(3): 225-229
- Cyprich, D., M. Kiefer & V. S. Lobačev - Fleas (Siphonaptera) of *Citellus musicus* Men. ground squirrel nests with emphasis on taxonomy and variability of *Citellophilus tesquorum ciscaucasicus* (Ioff, 1936). Biológia (Bratislava) 41(10): 999-1006
- Davies, T. H. - A checklist of the Arthropoda from Hawkes Bay, New Zealand. A Royal Forest & Bird Protection Society 1986 Community Service Project (ISBN-0-9597663-0-8), 81 pp. [Siph.: 58]
- Drucker, B. J. & W. B. Kottkamp - Siphonaptera and the sanitarian: An overview of the flea and recommendations for its control. J. environ. Hlth. 49(3): 154-157
- Dudich, A. - Blchy (Siphonaptera, Insecta) mikromamálií zo zemných pasci (Materiály k databanke fauny Slovenska). Zborn. Slov. nár. Múz., Prír. Vedy 32: 131-145 [German summary]
- Dudich, A. - Fleas (Siphonaptera, Insecta) - Ectoparasites of small mammals of the protected landscape territory of Muránska Planina. Ochrana Prirody 7: 151-168
- Dudich, A. - To the knowledge of ectoparasites of small mammals of Lúčanská Malá Fatra Mts. and Žilinska hollow (West Carpathians). Siphonaptera. In: 20th Camp of Nature Guardians - Review of scientific results. (Kunerad 14-22 July 1984: Slovakia), 66-80 [In Czech., no summary]

(1986)

- Dudich, A. & A. Stollmann - Share of the research station at the Institute of Experimental Biology and Ecology of the Slovak Academy of Sciences in Staré Hory on the teriological-parasitological documentation of protected territories of the Slovak Socialist Republic. Ochrana Prirody 7: 399-414
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It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).

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THE SYSTEMATICS ASSOCIATION SYMPOSIUM on the Biosystematics of Haemaphysagous Insects will be held at the Liverpool School of Tropical Medicine, Pembroke Place, Liverpool L3 5QA, England, from June 29 to July 2, 1987. Two presentations dealing with fleas are scheduled. These are:

"Insect tissue via the flea" by Miriam Rothschild and

"The coevolution of Siphonaptera (*Ceratophyllus celsus*) and Hirundinidae (*Hirundo pyrrhonota*)" by Cluff Hopla.

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